📊 Movie Rating Analysis – EDA Project (Data Science Internship)

📌 Overview

This project performs Exploratory Data Analysis (EDA) on a movie dataset to uncover patterns in:

1. Audience ratings
2. Financial performance
3. Genre trends
4. Director influence
5. Year-wise production growth

The analysis uses Python along with popular data-science libraries to visualize insights and discover meaningful relationships within the data.

🎯 Project Objective

To understand the key factors influencing movie success, including ratings, budget, box-office performance, and industry trends.

🧩 Problem Statement

Movies perform differently based on multiple factors such as budget, genre, production year, and directors.This project aims to identify which features significantly impact movie ratings and box-office success.

🛠️ Tools & Technologies Used

1. Python
2. Pandas
3. NumPy
4. Matplotlib
5. Seaborn
6. Jupyter Notebook / Google Colab

📁 Dataset Features

The dataset contains the following attributes:

1. Rating
2. Budget
3. Runtime
4. Writers
5. Genre
6. Year of Release
7. Director
8. Box Office Collection

The dataset was cleaned and pre-processed before visualization.

🔍 Analysis Performed

1) Univariate Analysis
   - Rating Distribution → Most movies have moderate ratings.

- Genre Distribution → Few genres dominate production.
- Year-wise Releases → Movie production increased over time.
- Top 10 Directors → Highlights filmmakers with the highest output.

2) Bivariate Analysis
   - Budget vs Box Office → Strong positive correlation.
   - Average Rating Year-wise → Ratings remain stable; spike seen in 1960–1980.
3) Multivariate Analysis
   - Correlation Heatmap
   - Strong correlation: Budget ↔ Box Office
   - Weak correlation: Runtime, Rating, Year

📌 Key Insights

- Dominant genres attract consistent investment.
- Movie production has grown significantly over decades.
- Higher budgets generally lead to higher box-office revenue.
- Audience ratings remain stable across years.
- Only budget strongly predicts financial success.
- Top directors significantly influence production volume.

📝 Conclusion

The EDA reveals predictable patterns in movie creation and performance:

- The industry is expanding steadily.
- Budget plays the biggest role in commercial success.
- Genre popularity and director contribution shape production trends.
- Audience expectations have remained consistent over time.

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